

Generation Interconnection Combined Feasibility / System Impact Study Report

For Queue Project Q14-0310

Kapstone 138 kV
November 28, 2015

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q14-0310. The Interconnection Customer proposes a 959.8 MW natural gas combined cycle facility interconnecting at the Kapstone 138 kV substation in the Virginia area. The request was received on February 20, 2014. The study evaluated system intact and single contingency conditions using the regional planning case, including 166 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 22 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$111,390,000.

3. Network Impacts

3.1 Generator Deliverability and Single Contingency Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Pratt'S to Innovative Solar 31 138.0 kV	Grimes to Louisa Generation acility 138.0 kV	340.9	247	63.2
Innovative Solar 31 to Kapstone 138.0 kV	Louisa Generation Facility to Grimes 138.0 kV	340.6	249	63.2
Grimes to Louisa Generation Facility 138.0 kV	Innovative Solar 31 to Kapstone 138.0 kV L	262.3	388	73.9
Kapstone Substation to Gimes 138.0 kV	Innovative Solar 31 to Kapstone 138.0 kV	195.0	399	74.2
Mustang to Pratt'S 138.0 kV	Loisa Generation Facility to Grimes Substation 138.0 kV	194.6	278	40.2
Nichols to St Lois 138.0 kV	Louisa Generation Facility to Grimes 138.0 kV	162.1	185	36.6
S Belleville to Nichols 138.0 kV line	Louisa Generation Facility to Grimes 138.0 kV	155.2	197	36.6
Northpoint to S Belleville 138.0 kV	Grimes to Louisa Generation Facility 138.0 kV Line	150.8	208	36.6
Mitchell Gardens to Kapstone 138.0 kV	Louisa Generation Facility to Grimes 138.0 kV Line	149.9	227	36.6
Northpoint to Mitchell Gardens 138.0 kV Line	Grimes to Louisa Generation Facility 138.0 kV	148.3	222	36.6
Louisa Generation Facility 345/138kV Autotransformer	Fluvanna PS to Louisa Generation Facility 345.0 kV Line	145.3	601	34.1
Pratt'S to Louisa Generation Facility 138.0 kV	Louisa Generation Facility to Grimes 138.0 kV	118.8	243	23.0

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
Gordonsville Energy Lp to Littlefork Substation 138.0 kV	Luisa Generation Facility 345 kV/138 kV Transformer Bank	115.9	266	27.5
Riverton 230 Substation to Remington Ct 138.0 kV line	Tuckasegee Tie to Merced 138.0 kV	115.3	377	8.4
Purdy to Chatom Substation 138.0 kV Line	Louisa Generation Facility to Grimes 138.0 kV line	111.1	279	6.2
Wheeler to Firebantz 138.0 kV	S Jordan to Chafee Gas Recovery Substation 138.0 kV	110.7	274	6.3
Spring Gap Tap to Mustang 138.0 kV	Louisa Generation Facility to Grimes 138.0 kV	108.4	217	16.7
Louisa Generation Facility to North Anna 345.0 kV	Louisa Generation Facility to Carrabelle 345.0 kV Line	106.4	1,686	25.1
Gordonsville Energy Lp to Louisa Generation Facility 138.0 kV Cir. 1	Louisa Generation Facility 345/138kV Transformer Bank	104.6	314	24.2
Gordonsville Energy Lp to Louisa Generation Facility 138.0 kV Circuit 2	Louisa Generation Facility 345-138 kV Transformer	104.6	314	24.2
Riverton 230 to Electric Avenue Facility 138.0 kV	Merced to Tuckasegee Tie 138.0 kV L	102.9	509	8.1
Lewisburg to South Jordan 138.0 kV line	Mount Pleasant to Middletown Jct 138.0 kV	100.7	246	5.0

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Innovative Solar 31 to Pratt'S 138.0 kV	Rebuild line with higher-capacity conductor	\$5,969,000	99% (\$5,920,000)
Innovative Solar 31 to Kapstone 138.0 kV	Rebuild line with higher-capacity conductor	\$10,761,000	99% (\$10,673,000)
Grimes to Louisa Generation Facility 138.0 kV	Rebuild line with higher-capacity conductor	\$2,592,000	94% (\$2,426,000)
Kapstone to Grimes 138.0 kV	Rebuild line with higher-capacity conductor	\$13,923,000	93% (\$12,994,000)
Mustang to Pratt'S 138.0 kV	Rebuild line with higher-capacity conductor	\$7,443,000	84% (\$6,229,000)
Nichols to St Lois 138.0 kV	Rebuild line with higher-capacity conductor	\$5,632,000	98% (\$5,549,000)
S Belleville to Nichols 138.0 kV line	Rebuild line with higher-capacity conductor	\$7,419,000	98% (\$7,310,000)
Northpoint to S Belleville Substation 138.0 kV	Rebuild line with higher-capacity conductor	\$7,657,000	98% (\$7,545,000)
Mitchell Gardens Substation to Kapstone Substation 138.0 kV Line	Rebuild line with higher-capacity conductor	\$3,907,000	98% (\$3,850,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
Northpoint to Mitchell Gardens 138.0 kV Line	Rebuild line with higher-capacity conductor	\$6,938,000	98% (\$6,837,000)
Louisa Generation Facility 345/138kV Autotransformer	Replace transformer with higher-rated unit	\$8,675,000	54% (\$4,648,000)
Pratt'S to Louisa Generation Facility 138.0 kV	Rebuild line with higher-capacity conductor	\$16,301,000	100% (\$16,301,000)
Gordonsville Energy Lp to Littlefork Substation 138.0 kV	Rebuild line with higher-capacity conductor	\$2,746,000	85% (\$2,325,000)
Remington Ct to Riverton 230 138.0 kV L	Rebuild line with higher-capacity conductor	\$19,007,000	22% (\$4,239,000)
Purdy to Chatom Substation 138.0 kV Line	Rebuild line with higher-capacity conductor	\$5,009,000	50% (\$2,501,000)
Wheeler to Firebantz 138.0 kV	Rebuild line with higher-capacity conductor	\$5,327,000	34% (\$1,800,000)
Spring Gap Tap to Mustang 138.0 kV	Reconductor line	\$4,199,000	77% (\$3,221,000)
Louisa Generation Facility to North Anna 345.0 kV	Reconductor line	\$16,549,000	23% (\$3,847,000)
Louisa Generation Facility to Gordonsville Energy Lp 138.0 kV Ckt 1	Reconductor line	\$695,000	100% (\$695,000)
Gordonsville Energy Lp to Louisa Generation Facility 138.0 kV Circuit 2	Reconductor line	\$705,000	100% (\$705,000)
Riverton 230 to Electric Avenue Facility 138.0 kV	Reconductor line	\$1,561,000	18% (\$277,000)
S Jordan to Lewisburg 138.0 kV line	Reconductor line	\$1,498,000	100% (\$1,498,000)
	Total allocated to project		\$111,390,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.